Selim El Fata

AI Engineering Student · Software Developer

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Fourth-year AI Engineering student who designs, builds and ships real software. I delivered a clinic management app that a dentist in Ghana now uses every day, and built a real-time black hole simulator that ray traces light through curved spacetime on the GPU. Open to freelance projects and full-time or part-time roles.

EXPERIENCE

Freelance Software Developer · Pro-denticare, dental clinic, Ghana — 2026

- Designed and built a desktop clinic management app, delivered to the client and in daily use: patient records, appointment calendar, treatment records, invoices, receipts, prescriptions and patient email.
- Built an interactive 3D dental chart in Three.js with an X-ray view, panoramic mode and a playable treatment-history timeline.
- Made it work fully offline with a local SQLite database, scheduled backups and migrations tested against every past version; packaged installers for macOS and Windows and shipped v1.2.
- Designed and built the product's marketing website.

Electron · React · TypeScript · SQLite · Three.js · Tailwind · Zustand · Zod

Software Development Intern · Melanion SAL — Jun 2026

- Completed a 40-hour software development program covering C# and object-oriented design, Git collaborative workflows, REST APIs and JSON, SQL data modelling and queries, and data cleaning, analysis and visualisation.

PROJECTS

Event Horizon · real-time black hole simulator — selim7417.github.io/event-horizon

- Wrote a GPU ray tracer for Schwarzschild spacetime from scratch, following every pixel's light ray backwards in real time; the black hole's shadow matches theory to within 0.35%.
- Physically based colour from Planck's law and CIE 1931, with Doppler beaming and gravitational redshift; dynamic resolution, temporal anti-aliasing and an eight-chapter guided tour.

WebGL2 · GLSL · JavaScript · no libraries

Personal Portfolio · animated portfolio website — selim7417.github.io

- Designed and built an animated, responsive portfolio with an interactive particle canvas and scroll-driven motion, tuned to a steady 60 fps with no frameworks.

EDUCATION

Artificial Intelligence Engineering · ESIB, Université Saint-Joseph (USJ) — Expected 2028

Currently in 4th year of a 5-year engineering programme · Beirut, Lebanon

SKILLS

Languages	Python · C++ · C# · JavaScript · TypeScript · SQL · GLSL · HTML/CSS
Frameworks	React · Electron · Three.js · WebGL2 · Tailwind CSS · Zustand · Zod
Tools	Git & GitHub · SQLite · REST APIs · Vite · Object-oriented design & design patterns
Spoken	Arabic (excellent) · English (excellent) · French (very good)

STRENGTHS

End-to-end ownership
Took a client's app from first sketch to daily use, including design, code, testing and delivery.

Detail-oriented
Tested database migrations against every past version of the clinic app before each release, so updates never break existing data.

Fast learner
Taught myself GPU programming and 3D graphics to build a physics-accurate simulator from scratch.

Clear communicator
Explains technical work in plain words and works directly with clients to understand what they need.